DOC 330: Designing Innovative Solutions
Tutorial 4 – Idea Validation

Activity

1. **Decide the problem.**

   Problem - The healthcare industry lacks a personalized wellness platform and struggles with convenient access to healthcare providers.

   Solutions - We propose an innovative AI website that integrates advanced technologies, personalized health recommendations, and secure database management to empower individuals in their wellness journey and bridge the patient-doctor gap.

2. **Identify the scope (size) of the problem.**

   The healthcare industry faces a significant challenge in providing comprehensive and personalized wellness management for individuals. Existing systems often lack the necessary tools and platforms to empower individuals to take an active role in their well-being, access tailored recommendations, and effectively communicate with healthcare providers. This lack of a holistic solution hampers the delivery of personalized care and limits the potential for optimal wellness outcomes.

   Firstly, there is a need for a comprehensive platform that enables individuals to manage their wellness in a personalized manner. Current healthcare systems often lack the ability to consider individual medical history, preferences, and lifestyle choices when providing recommendations or treatment plans. A solution is required to bridge this gap and provide tailored approaches that cater to the unique needs of everyone.

   Secondly, there is a lack of health records and other relevant health data in a centralized and secure manner. This fragmentation of health information makes it difficult for healthcare professionals to access and utilize comprehensive patient data for efficient decision-making. A solution is needed to establish a robust database management system that securely stores and manages health records, allowing for seamless access and analysis of patient information.

   In summary, the problem lies in the absence of a comprehensive and personalized platform that enables individuals to manage their wellness, access tailored recommendations, and effectively interact with healthcare providers. This problem encompasses the need for personalized wellness management, integration of electronic health records, and improved communication channels. Addressing this problem will lead to enhanced individual empowerment, improved healthcare decision-making, and ultimately, better wellness outcomes.

- Technology Scope:

- Development of an AI-enabled website with advanced algorithms for personalized health recommendations.
- Integration of secure and comprehensive database management systems.
- Implementation of user-friendly interfaces and interactive features.

- Healthcare Scope:

- Personalized wellness management approaches that consider individual medical history, preferences, and lifestyle choices.
- Integration of health records for efficient healthcare decision-making.
- Facilitation of effective communication and interaction between patients & doctors.

- User Engagement Scope:

- Empowering individuals to actively participate in their wellness journey through access to personalized recommendations.
- Knowledge sharing and peer-to-peer support through interactive features such as blogging.

- Impact Scope:

- Improving overall wellness outcomes through personalized health recommendations and proactive management.
- Empowering individuals to take charge of their health by providing them with and resources.

3. Problem Justification or idea validation through

a. Literature survey

- **Limited Access to Healthcare Services**

  Sri Lanka faces the challenge of limited access to healthcare services, which hinders individuals from receiving timely and appropriate medical care. The country suffers from insufficient healthcare facilities and resources, particularly in rural areas, leading to long waiting times and delays in receiving care. Geographic barriers pose challenges in accessing healthcare services in remote and isolated regions, while financial constraints and lack of health insurance coverage prevent many from affording necessary healthcare. Additionally, the limited availability of specialized healthcare services and language barriers further impede access. The consequences include delayed diagnoses, compromised health outcomes, and exacerbated health disparities. Addressing these issues is crucial to ensuring equitable access to healthcare services for all Sri Lankans.

  Through the introduction of an AI-enabled website, the proposed system aims to address the problem of limited access to healthcare services by providing individuals with convenient and remote access to healthcare.

- **Access to Health Records**

  One of the problems in the healthcare system in Sri Lanka is the absence of a comprehensive and centralized patient database. Sri Lanka lacks a robust system for maintaining and managing patient health records, which creates challenges in delivering efficient and coordinated healthcare services. The absence of a centralized database leads to fragmented health information and difficulties in accessing complete patient records when needed. This hinders healthcare professionals' ability to make well-informed decisions, potentially resulting in delays, duplicated tests, and compromised patient care. Establishing a secure and comprehensive patient database is crucial to ensure seamless access to medical histories, improve healthcare coordination, enhance patient safety, and facilitate better healthcare outcomes in Sri Lanka.

  The proposed concept aims to address the problem of the absence of a patient database in Sri Lanka's healthcare system. By implementing an AI-based website, the concept seeks to establish and maintain a comprehensive and secure database of patient health records. This centralized database will securely store and manage patient information, including medical history, test results, diagnoses, and treatment plans. Healthcare providers will have seamless access to this information, enabling them to make well-informed decisions, provide efficient care, and avoid duplication of tests or treatments. By establishing a robust patient database, the proposed concept offers a solution to the problem.

- Lack of AI based health care systems

  While AI-based healthcare systems have shown significant progress, it is important to note that there is currently no instant solution that can address all patient problems. Healthcare is a complex and individualized field, and developing a comprehensive AI system requires extensive research, validation, and integration with existing practices. Moreover, considerations such as data privacy, security, and ethical concerns must be carefully addressed. Although AI holds great potential for transforming healthcare, it is an ongoing process, and further advancements are needed to fully harness its capabilities in effectively addressing diverse patient challenges.

  The proposed idea includes an innovative feature that allows users to engage in interactive communication with AI bots to receive medical assistance. By integrating chatbot functionality into the system, individuals can have real-time conversations with AI-powered virtual assistants. These chatbots can provide information, answer questions, and offer basic medical advice based on user input. This interactive feature enhances accessibility and convenience for individuals seeking healthcare assistance, as they can receive prompt and personalized responses to their queries or concerns.

References

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B. **gap analysis**

Features	Similar systems (Websites) Locally / Global						Proposed Solution (Idea)
	Sri Lanka Red cross Society	Lanka hospitals	National hospital Sri Lanka	National Health Service	Medicare (USA)	Sistema Único de Saúde (SUS)	
Interactive chat-bot	No	No	No	No	No	No	Yes
Virtual consultations	No	Yes	No	Yes	Yes	No	Yes
Health recommendations	Yes	No	No	Yes	No	Yes	Yes
Access to health records	No	No	No	Yes	Yes	No	Yes
Health Education information	No	Yes	No	Yes	Yes	Yes	Yes
Reminders & Notifications	No	Yes	No	No	No	No	Yes
Medication management	No	No	No	No	No	No	Yes
Symptom checker	No	No	No	No	No	No	Yes
Wellness programs	Yes	Yes	No	Yes	Yes	No	Yes
Language support	No	No	No	No	No	No	Yes
Emergency assistance	No	No	No	No	No	No	Yes